CHRISTOPHER MARONGA

Statistics, Data Management & R Programming

SKILLS

R	6+ yrs
REDCap	5+ yrs
SQL/MySQL	4+ yrs
GitHub	3+ yrs
LaTeX	2+ yrs
SAS	1+ yrs
ODK	1+ yrs

EDUCATION

2019 - 2021

MSc. in Statistical Science
Strathmore University

2009 - 2013

B.Sc. in Mathematics
Jomo Kenyatta University of Agriculture and Technology

2004 - 2007

K.C.S.E
Sawagongo High School

Kenya Certificate of Secondary Education

PROFILE

Experienced Data Analyst with demonstrated history of working in Health research Data Management and Statistical analyses.

Long run experience with data and model exploration tools, designing of data products in Rshiny and Rmarkdown and Co-organizer of NairobiR, R user group.

Knowledgeable in designing of database and data collection tools in RED-Cap, ODK and selected RDMS

WORK EXPERIENCE

R Programmer Jun2021 - Presently
University of Oxford

Working with the Health Economics and Outcome Research Group to develop and apply innovative algorithms/systems to extract knowledge and insights from health care data

- Mining, cleaning and processing of healthcare data using R
- Responsible for developing and maintaining common library of R functions for team collaboration
- Running statistical analysis and compiling of high-quality reports and presentations using Rmarkdown
- Support in developing study manuscripts for publication
- Contribute in developing joint R packages specific to health economics
- Build interactive web applications using R-Shiny for dynamic report dissemination
- Responsible for reviewing code written by other researchers, help troubleshoot and suggest efficient ways of coding

Technologies:

- Rshiny
- Rmarkdown

Achievements:

- Created a fully automated reports aggregation for a micro-simulation model using Rmarkdown

CONTACT

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Clinical Data Manager Oct 2017-May 2021

KEMRI-Wellcome Trust Research Programme,

The CHAIN Network is a large multi-continent, multi-country cohort study with a total of 9 sites across Africa and East Asia

- Coordination of data discrepancy checking, cleaning, validation, data retrieval, organizing, and transferring between systems across all the 9 sites
- Liaison with data management teams across all sites to oversee rapid resolution of data queries
- Generation of standardized quarterly and annual reports for sites' Principal Investigators and Network lead scientists
- Play a central role in ethical data sharing within the network researchers
- Data systems infrastructure maintenance and troubleshooting
- Perform data analysis

Technologies:

- Rshiny
- Tidyverse
- Rstudio Server
- Shiny Server
- Git for collaboration and documentation versioning

Achievements:

- Developed a robust Shiny dashboard for automated data cleaning and report output
- Boosted data managers productivity through hosting automated customized sites weekly reports on Shiny dashboard

Technical Advisor – Data Management Jan 2017-Sep 2017

Partners PrEP Scale-Up Project (KEMRI CCR Project)

An implementation project to scale-up delivery of antiretroviral-based HIV-1 prevention methods to serodiscordant couples in 24 public HIV-1 care centers in central and western Kenya

- Provide technical advise to HIV care clinics in all aspects of data management and collection, including regular trainings on data collection practices
- Prepare and submit to coordinating centre a cumulative report of selected clinics' indicators on a monthly basis for evaluation of project progress
- Conduct monthly conference with donors for update on study progress and advise on technical issues that may arise
- Capacity building of Project Research assistants and data staff on offsite clinics
- Monitor study accrual rates and generate QC reports for the 13 participating sites and follow up to ensure a timely resolution of the data queries

Technologies:

- SurveyCTO
- R software

Achievements:

- Harmonized and centralized key indicator reporting for clinics using surveyCTO and automatically aggregate the reports for consumption.

Assistant Data Manager Jul 2014-Jan 2017

Partners in Health, Research and Development

Enviroment of multi-Clinical Trials and Research studies and invovled in over 5 clinical trails data management activities

- Designing of REDCap databases for participants data storage
- Preparation of daily, weekly and monthly study data summaries to track and inform the trials' progress
- Perform data entry of participants' records into REDCap data base
- Configure the SCIP application on tablets for SCIP Msurvey trials
- Design of data capturing systems/tools, development of SOPs for data recording and entry, and case report forms for data collection.
- Data QA/QC review on REDCap, validation and discrepancy checking
- Perform regular cloud data backup for the study, develop comprehensive system logs in excel to track data query work flow
- Co-ordinate ordering/procuring and distribution of both hardware & software and keep up to date the site's inventory for IT equipment
- Maintain and update on a daily basis participant's master data base, which contains client's details, progress through the trial, adverse events and other relevant research related information
- Closely working with study monitor to oversee quality assurance and quality control procedures for the clinical trial
- Training and supervising of research assistants, interns and data entry clerks at the Thika site
- Maintain systematic data storage of participants' data (both electronic and binder files) as per Good Clinical Practice(GCP).

Technologies:

- Ms Access
- REDCap
- R for reporting and data management

Achievements:

- Developed a robust system using Ms Access and Ms Excel for automated patient follow up tracking and report generation on missed appointments and lost to follow up

- Design over 6 usable databases both in REDCap and Ms Access

Research Assistant(Intern) Oct 2013-Jun 2014
Partners in Health, Research and Development

A project to evaluate feasibility and potential impact of text messages on HIV awareness in young women in rural Kenya. Tasks included

- Study coordinator, strategic planning of participant recruitment into the study and short messages schedules with msurvey Kenya, performed data cleaning and preliminary data analysis and report generation using SPSS
- Designed storage data base and managed data entry, data cleaning and regular survey data back-up
- Assist t2t PI in drafting project proposal, abstracts, publications and budgets
- Responsible for confidential storage and documentation of study data and link logs according to GCP
- Preparations of bi-weekly and weekly data summaries as required for study retention and follow up to assess and inform status/progress of t2t study project
- Perform any duty as assigned by t2t Principal Investigator, Site coordinator and Lead Clinician.

Technologies:

- Ms Access
- SPSS for reporting

Achievements:

- Oversaw data management for the study from start to successful completion with a high quality data output

Abstracts and co-authored abstracts

Christopher Maronga, *Narshion Ngao, Eric Owino, Priya Sukhtankar,Moses Ngari*; **Innovative open source approaches to automating clinical data management for a large multi-centre cohort study(KASH 2019)**

*Njambi Njuguna, Kenneth Ngure, Nelly Mugo, Carole Sambu,**Christopher Sianyo**, Renee Heffron*; **Assessing feasibility of conducting a text message based HIV prevention Intervention among young women in rural Kenya (AIDS2014 conference, Melbourne Australia)**

*Elizabeth Irungu, Monisha Sharma, Renee Heffron, Nelly Mugo, Kenneth Ngure, Roger Ying, **Christopher Sianyo**, Connie Celum, Jared Baeten, Ruanne Barnabas*; **The incremental cost of delivering PrEP as a bridge to ART to HIV serodiscordant couples in public HIV care clinics in Kenya**.

Hobbies and Interests

- I love listening to music
- R computing at free time is my safe haven

References

Dr. Moses Ngari
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